# Predetermined Motion Time Systems


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The controversial step in direct time study is performance rating. It is based on a judgment by the time study analyst, and there is always room for disagreement between the worker doing the task and the analyst evaluating the worker's pace. An alternative to direct time study that does not include the performance-rating step, at least not directly, is the family of work measurement techniques known as predetermined motion time systems.

A ***predetermined motion time system*** (PMTS), also called a ***predetermined time system***, is a database of basic motion elements and their associated normal time values, together with a set of procedures for applying the data to analyze manual tasks and establish standard times for the tasks.[1] The PMTS database is most readily conceptualized as a set of tables listing time values that correspond to the basic motion elements, the lowest level in our hierarchy of manual work activity. They include motions such as reach, grasp, move, and release. The time required to perform these basic motions usually depends on certain work variables. For example, the time to reach for an

[1]The Work Measurement and Methods Standards Committee (ANSI Standard Z94.11-1989) defines a predetermined motion time system as follows: "An organized body of information, procedures, techniques, and motion times employed in the study and evaluation of manual work elements. The system is expressed in terms of the motions used, their general and specific nature, the conditions under which they occur, and their previously determined performance times."


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object increases with the distance of the reach, among other factors. The time to move an object depends on its weight and the distance it is moved. The normal time values in the database have been determined based on extensive research of manual work activity, usually consisting of frame-by-frame analysis of motion pictures of the activity.

In this chapter we provide a general discussion of predetermined motion time systems, which were first identified around 1925 (Historical Note 1). We then examine two important examples of these systems: methods-time measurement (MTM) and the Maynard Operation Sequence Technique (MOST).

### HISTORICAL NOTE 1 PREDETERMINED MOTION TIME SYSTEMS

The notion that manual work consists of basic motion elements is attributed to Frank B. Gilbreth, who was a pioneer in the subject of motion study. His studies resulted in the first methodical classification of motion elements, which he called ***therbligs*** (Gilbreth spelled backwards—well, almost). The 17 therbligs consist mostly of arm and hand motions.

Asa B. Segur is credited with developing the first commercial predetermined motion time system, called Motion-Time Analysis (MTA). Segur was aware of Gilbreth's classification scheme for basic motion elements, and MTA's 17 basic motion elements agree closely with Gilbreth's 17 therbligs. Segur began the development of MTA in 1922 by analyzing motion picture films taken during World War I of operators performing factory tasks. The films were originally intended for use as training materials for blind and otherwise handicapped workers, but Segur constructed his system of motions and times from them. Supposedly for commercial reasons, he limited public access to MTA, and this is one of the reasons why his system is no longer used. However, his work influenced those who followed him in developing other predetermined motion time systems.

One of these analysts was Joseph H. Quick, who developed the Work-Factor system between 1934 and 1938. The system was based on the analysis of large numbers of motion picture films, snapshots, stroboscopic lighting techniques, and stopwatch studies taken of many different kinds of industrial operations. Among Quick's contributions in PMTS research were his studies involving cognitive work, such as visual inspection and similar technical labor.

Harold B. Maynard stands as one of the more important figures in the development of predetermined motion time systems. He is largely responsible for the Methods-Time Measurement (MTM) system, and the Pittsburgh-based consulting firm of H. B. Maynard and Company bears his name. In 1946, Maynard began development of MTM with colleagues G. J. Stegemerten and J. L. Schwab under the sponsorship of the Westinghouse Electric Company. Their study began by analyzing movies of drilling operations and relating the work patterns to therbligs. The therblig classification was found to be wanting in some respects, and many of the basic motion elements were revised, renamed, or removed. MTM became the most successful and widely used first level PMTS after its release in 1948. The MTM database of motion elements was used

in the development of many higher-level systems, including those in the MTM family (see Table 5 later in this chapter).

Other individuals who should be mentioned for their efforts in developing PMT systems include G. B. Bailey and R. Presgrave, who developed Basic Motion Timestudy in 1950; R. M. Crossan and H. W. Nance, who developed Master Standard Data in the late 1950s; and G. C. Heyde, who developed the Modular Arrangement of Predetermined Time Standards (MODAPT) in the mid 1960s.

Kjell B. Zandin is largely responsible for the development of the Maynard Operation Sequence Technique (MOST). Starting in the late 1960s, he led a group at the Swedish Division of H. B. Maynard & Company in studying applications of MTM. It was observed that there were similarities in the motion patterns and MTM sequences of many manual operations. This ultimately resulted in the definition of three principal motion groups, which were called sequence models in the MOST system. MOST was first introduced in Sweden in 1972 and in the United States in 1974. Today, it is one of the most widely used PMT systems.

Subsequent further development of predetermined motion time systems has been in the computerization of several of these systems as commercial products. Examples include MOST for Windows and MTM-LINK.

## 1 OVERVIEW OF PREDETERMINED MOTION TIME SYSTEMS

This section provides a general introduction to PMT systems. First, we discuss the usual procedure in applying a predetermined motion time system to a task. One of the significant advantages of a PMTS is that it can be used to determine a standard time for a task before that task is in production. It can also be used to set a standard for an existing task. We then indicate the differences among PMT systems and how they can be classified.

### 1.1 The PMTS Procedure

All PMT systems are applied using the same general procedure. The differences are in the details. To apply a predetermined motion time system to a task, the analyst must accomplish the following steps:

1. Synthesize the method that would be used by a worker to perform the task (or analyze the method that is being used by a worker in an existing task). The method is described in terms of the basic motions comprising the task, based on a defined workplace layout and set of tools (if tools are used).[2]
2. Retrieve the normal time value for each motion element, based on the work variables and conditions under which the element will be (is) performed. Sum the normal times for all motion elements to determine the normal time for the task.

[2]Different predetermined motion time systems have different basic motion definitions, and the analyst must describe the method using the definitions and conventions of the particular system.

3. Evaluate the method to make improvements by eliminating motions, reducing distances, introducing special tools, using simultaneous right and left hand motions, and so on. The objective of this evaluation is to reduce the normal time. The evaluation process is facilitated by the detailed listing of basic motion elements and corresponding time values.
4. Apply allowances to the normal time to determine the standard time for the task.

If the emphasis in the application is methods improvement, then step 4 may be omitted. If the emphasis is setting a time standard, then step 4 is required.

The basic time values in PMT systems do not include any allowances, so the user organization must add an allowance for personal time, fatigue, and delays according to its policies on allowances. This is step 4 in our PMTS procedure outlined above. The following equation is the usual way in which allowances are applied to determine the time standard for a task:

$$T_{std} = T_n(1 + A_{pfd}) \tag{1}$$

where $T_{std}$ = standard time, min; $T_n$ = normal time, which is the sum of the basic motion time values that make up the task (step 2 with possible adjustments resulting from step 3 in the PMTS procedure), min; and $A_{pfd}$ = PFD allowance factor.

A performance rating step is not necessary in a predetermined motion time system, because the basic motion times are already rated at standard (100%) performance. However, different PMT systems use different definitions of standard performance. MTM and MOST use the definition of "daywork" standard performance. If the organization using either of these systems defines standard performance differently, then a universal adjustment must be made to all time values in the database to bring these data into agreement with the organization's definition.

Similarly, for predetermined motion time systems other than MTM or MOST, an adjustment in all data values may be necessary to achieve agreement with the organization's definition of standard performance. For example, Motion Time Analysis (MTA) time values were determined based on a faster working pace, and therefore the basic motion times are less than MTM times by about 20%. Therefore, if a company were to use the MTA system but wanted to set standards using the "daywork" as the standard performance level, then all of the MTA times must be multiplied by 125%.

### 1.2 PMTS Levels and Generations

There is a variety of predetermined motion time systems. Although we might think that the basic motions of the human body should be the same in all PMT systems, some PMT systems combine basic motions into larger motion elements. Thus, we have various levels of PMT systems. First-level systems use the basic motion elements (e.g., reach, grasp), while higher-level systems combine several basic motion elements into ***motion aggregates***, combined motion sequences that are commonly used in work situations. For example, the basic motions ***reach*** and ***grasp*** might be combined into one element called

***get*** for use in a higher-level PMT system. This motion aggregate would be common in assembly work, for example.

First-level PMT systems tend to be very detailed, with body motions differentiated very precisely in their databases. For example, ***reach*** and ***grasp*** are typically distinguished as two separate motion elements, even though they are commonly carried out as one smooth arm and hand movement. The times for the basic motions tend to be very short, sometimes much less than a second. Higher-level systems use condensed databases, with fewer body motions contained in the tables and longer time values for each motion sequence. The tables are simplified, with fewer work variables that must be identified about the task by the analyst.

Chronologically, first-level PMT systems were the first to be developed, and then second- and higher-level systems were subsequently constructed based on the first-level systems. Because of this chronological development of the systems, the level of the system usually corresponds to the ***generation*** of the system. First-level PMT systems are called first generation systems, and the subsequent systems are second and third generations. For example, MTM-1 is first generation. MTM-2 is second generation and is based on MTM-1. MTM-3 is a third generation MTM system.

Methods descriptions in first-level systems consist of long lists of elements, and the time to determine a standard for the task is long (high application speed ratio). For example, using MTM-1, a first-level system, the time to set a standard is a factor of about 250 compared to the duration of the task time [7]. For a task of 1 minute, it takes over 4 hours of analyst time to establish the standard time. Higher-level PMT systems require less application time due to their use of elements that combine basic motions. For example, MTM-2 is a higher-level system that is based on MTM-1, and the time to set a standard is a factor of 100 compared to the task time. The cost of this greater convenience is that the accuracy of higher-level systems tends to be less than for a first-level system. The general characteristics of PMT system levels are summarized in Table 1.

More than 50 predetermined motion time systems have been developed over the years. Most of these systems are no longer used. Table 2 provides a brief description of the major PMT systems that were developed, with an emphasis on those that are commercially important today. The most widely used systems today are based on MTM. Section 2 discusses MTM, both the first-level system (MTM-1) and some of its higher-level versions. Section 3 discusses MOST, a widely used higher-level PMTS based on MTM.

**TABLE 1** Characteristics of PMT System Levels

| Characteristic | First-Level PMTS | Higher-Level PMTS |
|---|---|---|
| Accuracy | Most accurate | Less accurate |
| Application time | Much time to set standard | Less time to set standard |
| Suited to specific types of tasks | Highly repetitive | Repetitive or batch |
| Cycle times | Short cycle (e.g., 1 min) | Longer cycle times feasible |
| Motion elements | Basic motions | Aggregates of basic motions |
| Methods description | Very detailed | Less detailed, easier to apply |
| Flexibility of application | Highest flexibility | Less flexibility |

**TABLE 2** Summary of Predetermined Motion Time Systems

***Motion-Time Analysis (MTA)***: A first-level system developed by A. B. Segur in the mid- and late 1920s. Its primary importance today stems from the fact that it was the first PMTS to be developed and applied. Today it is seldom used if at all. There are 17 basic motion elements in MTA, involving one or more body movements. The elements are classified as positive (considered useful to the task), negative (motions that should be reduced or eliminated), or either (might be positive or negative).

***Work-Factor (WF)***: A first-level PMTS developed by J. H. Quick, W. J. Shea, and R. E. Koehler in the mid-1930s, with applications starting in 1938. It is still used today and at least one simplified version has been developed to reduce application time. Coverage includes both factory and office work.

***Methods-Time Measurement (MTM)***: Developed by H. B. Maynard, G. J. Stegemerten, and J. L. Schwab in the mid-1940s. The original MTM was a generic, first-level system published as a book in 1948. Today, that original PMTS is known as MTM-1, and several higher-level and specific PMT systems have been developed based on MTM-1. We describe the MTM family in Section 2.

***Basic Motion Timestudy (BMT)***: Developed by R. Presgrave, G. B. Bailey, and J. A. Lowden in the early 1950s. Its initial applications were in 1952, and it was published in book form in 1958. It is considered a first-level system, but some of its motion elements are combinations of therbligs. At least one second-level PMTS has been developed based on BMT.

***Master Standard Data (MSD)***: One of the first second-level systems. It is based on MTM-1 and was developed in Canada by R. M. Crossan and H. W. Nance in the late 1950s (published as a book in 1962) to reduce the analyst's time needed to set a time standard for manual production operations with fewer than 100,000 cycles per year. Its coverage emphasizes reach-grasp-move-release motions that are common in such operations.

***Maynard Operation Sequence Technique (MOST)***: Developed by K. B. Zandin and others at the Swedish Division of H. B. Maynard Company as a higher-level PTMS based on MTM-1 and MTM-2. Its coverage emphasizes production operations and material handling. We describe MOST in Section 3.

***Modular Arrangement of Predetermined Time Standards (MODAPTS)***: A second-level PTMS developed by C. Heyde and others in Australia during the mid-1960s. It is based on MSD, MTM-1, and MTM-2. The basic motion element in MODAPTS is a finger movement, and other body motions are expressed in terms of this element. Coverage includes production operations and material handling, with a special version covering office work.

*Source:* Compiled from [1], [7], and other sources.

## 2 METHODS-TIME MEASUREMENT

Methods-Time Measurement (MTM) is a PMTS that can be used to analyze the method for performing a given task and to set a time standard for the task. MTM is defined by its developers as follows [4]:

> Methods-Time Measurement is a procedure which analyzes any manual operation or method into the basic motions required to perform it and assigns to each motion a predetermined time standard which is determined by the nature of the motion and the conditions under which it is made.

The hyphenation of "Methods-Time" was intended by its developers to emphasize the important connection between the basic motions used to perform a task and the time values associated with these motions. MTM is an analysis tool by which the method for a task is divided into its component elements to determine if any methods improvements can be made (step 3 in the PMTS procedure). Only after this analysis should a standard time be established.

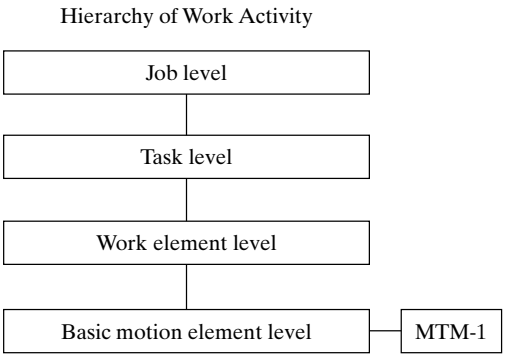



**Figure 1** The position of MTM motion elements in our work hierarchy.

MTM is actually a family of predetermined motion time systems. The original MTM is now called MTM-1. It is the first-level PMTS upon which all of the other MTM systems are based. We describe MTM-1 in Section 2.1. Other versions of MTM are briefly discussed in Section 2.2. Training in the members of the MTM PMTS family is available through the MTM Association, located in Des Plaines, Illinois.[3]

### 2.1 MTM-1

In our hierarchy of work activity, MTM-1 operates at the basic motion element level, as illustrated in Figure 1. Most of the MTM-1 basic motions involve hand and arm movements, although elements are also provided for eye, leg, foot, and body actions. Many of the basic motions correspond to the original therbligs as developed by Frank Gilbreth. However, the definitions of these basic motions have been adjusted to conform to the needs of MTM, and new elements not included among the original therbligs have been added.

Time units in MTM are called TMUs (time measurement units). MTM was developed by studying motion pictures of work activity, and the time units for MTM were originally defined as the time per frame of motion picture film. This proved awkward, and the TMU is now defined as follows:

$$1 \text{ TMU} = 0.00001 \text{ hr} = 0.0006 \text{ min} = 0.036 \text{ sec}$$

Conversely, there are 100,000 TMUs in 1 hr, 1667 TMUs in 1 min, and 27.8 TMUs in 1 sec.

Table 3 defines the MTM-1 motion elements, and Table 4 presents a tabulation of their time values. As with other PMT systems, the analyst must describe the task in terms of these basic motion elements, taking note of the work variables that influence the element time and accounting for simultaneous right-hand and left-hand motions,

[3]MTM Association, 1111 East Touhy Avenue, Des Plaines, IL 60018; their Web site can be accessed at www.mtm.org.

**TABLE 3** Summary of MTM-1 Motion Elements

Motion Element (Symbol) and Description

**Reach (R)**: A basic motion element involving movement of the hand or fingers. Its purpose is to move the hand or finger to a new destination. The corresponding therblig is "transport empty." If the hand is holding something during the motion, it is still classified as a reach if its primary purpose is to reposition the hand or fingers and not to move the object—for example, reaching for an object while holding a cigarette. Reach depends on two work variables: (1) the distance moved and (2) the case under which the reach is performed. The case refers to such factors as whether the reach is toward an object in a known location or the object is jumbled amongst other objects and must be searched out. The five cases are identified in Table 4 (a), which provides time values as a function of distance and case. If the hand is in motion before and/or after the reach, then the time is reduced slightly, as indicated in the table under "hand in motion" for cases A and B. The complete symbol for a given reach element in MTM-1 includes the distance and the case (e.g., R10C).

**Grasp (G)**: A motion element used by the fingers and hand to gain control of one or more objects. It is commonly employed as a prerequisite motion for performing the next basic motion, which is likely to be a move—for example, grasping an object prior to moving it. There are five categories of grasp: (1) pickup, (2) regrasp, (3) transfer, (4) select, and (5) contact. Within the pickup and select categories, there are several cases that must be distinguished. Table 4 (b) defines the various categories and cases and lists the times for them. The complete symbol for a grasp includes category and case (e.g., G1C3).

**Move (M)**: A hand or finger motion whose primary purpose is to relocate an object. The corresponding therblig is "transport loaded." The move element includes pushing or sliding an object across a surface, so long as the hand controls the object. Move depends on three work variables: (1) the weight of the object being moved, (2) the distance of the move, and (3) the case under which the move is performed. There are three cases, as described in Table 4 (c). The weight variable is divided into classes, and the user selects the next higher weight given in the table for the object weight in the given application. The complete symbol for move includes the distance, case, and weight—for example, M6B10 symbolizes a move of 6 in. under case B for an object weight of 10 lb. The following formula must be used to determine the TMU value from Table 4 (c):

$$\text{TMU} = \text{Constant} + \text{Factor} \times (\text{TMU value from table})$$

where Constant is the constant value from the table for the object weight in the application, TMU; Factor is the factor value from the table for the object weight; and (TMU value from table) is the TMU value from the table for the corresponding distance of the move and the move case. For example, given the symbol M6B10, the formula would be used as follows:

$$\text{TMU} = 3.9 + 1.11(8.9) = 13.8 \text{ TMU (about } {}^{1}\!/_{2} \text{ sec)}$$

**Position (P)**: A relatively short hand motion (no greater than 1 in.) employed to align, orient, or engage one object relative to another. It is usually preceded by a move. ***Align*** refers to the relative positioning of longitudinal axes of the objects; ***orient*** refers to the relative positioning of rotational axes of the objects; and ***engage*** refers to the insertion of one object into the other following either an alignment or an orientation. Accordingly, the position element may occur several times together depending on the number of separate positioning moves required. The work variables that affect the time for a position motion are the following: (1) pressure required to achieve the fit, (2) symmetry of the objects, and (3) ease of handling. Pressure to achieve fit is divided into three categories: loose (class 1), close (class 2), and exact (class 3). Symmetry is also divided into three categories: symmetrical (S)—for example, a round peg in a round hole; nonsymmetrical (NS)—for example, a key inserted into a lock; and semisymmetrical (SS), which is any case between S and NS. Ease of handling is simply divided into easy (E) and difficult (D). The categories are listed in Table 4 (d) along with the time to perform the element. The complete symbol for a position includes the three work variables (e.g., P3NSD).

**Release (RL)**: A hand and finger motion element whose purpose is to surrender control of an object. There are only two cases of release: (1) the fingers open to effect the release and (2) contact release with no finger motion. The complete symbol for release identifies the case (e.g., RL1 or RL2). Times are given in Table 4 (e).

**Disengage (D)**: A hand and finger motion that causes the separation of two objects from one another, where the objects were previously held together by some force. Thus, disengagement breaks the force, resulting in a recoil action by the hand. The work variables that affect the time of a disengage motion are (1) class of fit and (2) ease of handling. Classes

(*continued*)

**TABLE 3** (*continued*)

| Motion Element (Symbol) and Description |
|---|
| of fit are loose (slight effort required, class 1), close (normal effort, class 2), and tight (considerable effort, class 3). Ease of handling is either easy (E) or difficult (D). The disengage symbol indicates the two variables (e.g., D3E). Times are given in Table 4 (f). |
| **Turn (T)**: A basic motion element that involves rotation of the hand and wrist about the long axis of the forearm. The hand can either be holding an object (e.g., turning a dial on a machine) or empty. Turn depends on two work variables: (1) the degrees of turn and (2) resistance to the turn. Table 4 (g) lists the time values as a function of these factors. The maximum turn in the table is 180°, which is imposed by the anatomical structure of the elbow and forearm. Resistance to turn is indicated by force of resistance, divided into three categories in the table. The complete symbol for turn includes both variables; for example, T90L symbolizes a turn of 90° and a large resisting force. |
| **Apply pressure (AP)**: An MTM-1 element that involves the application of force, but the force results in little or no movement. The force is usually applied by the hand and/or fingers, but the element applies to a force applied by any body member. There are two categories of apply pressure: (1) apply pressure alone (APA) and (2) apply pressure preceded by a regrasp (APB). Time values for these two cases are given in Table 4 (h). |
| **Eye travel (ET)**, **eye focus (EF)**, and **reading**: Basic eye usage motions that are important enablers for performing manual work. However, the effect of the eyes during a motion element is usually accounted for in the tabulated TMU values. For example, a reach motion that requires greater hand-eye coordination takes longer; for example, compare Case C and Case A in Table 4 (a). On the other hand, there are situations when eye travel and/or eye focus are prerequisites for subsequent motions. In these cases, they must be counted as separate basic elements. Times for the two basic eye usage motions are given in Table 4 (i). In addition, MTM-1 uses a normal time for reading text as 5.05 TMUs per word. Thus, reading a 200 word paragraph would be allowed 1010 TMUs (about 36 sec). |
| **Body, leg, and foot motions**: Additional basic motions that involve moving the body, one or both legs, and/or one or both feet. The motions include walking, bending, stooping, standing from a seated position, and sitting from a standing position. These activities are distinct from those performed at a workbench, most likely with the worker in a sitting position. These elements and others are identified in Table 4 (j), together with their symbols and normal times. |
| **Simultaneous motions**: Basic motion elements that are performed at the same time. In general, it is desirable for more than one body member to be moving simultaneously. For example, it is desirable for both the right hand and left hand to be active during the task, not only sequentially but also simultaneously. When basic motion elements can be combined simultaneously, the time required for the combination is the greater of the two parallel motions. This is called the "limiting principle," which was first proposed by the original developers of MTM (Maynard, Stegemerten, and Schwab, Historical Note 1). Certain combinations of motions are more difficult to perform simultaneously than others. The degrees of difficulty can be classified as follows: (1) easy to perform simultaneously, (2) can be performed simultaneously with practice, and (3) difficult to perform simultaneously. Table 4 (k) indicates the degree of difficulty for various combinations of basic motion elements. In degrees of difficulty (1) and (2), the limiting principle applies, assuming sufficient practice is allowed for the worker in (2). In degree of difficulty (3), both times should be added. |

internal elements when the task involves operation of a machine, and other special circumstances. The analyst then retrieves the time for each basic motion element and sums the times to obtain the normal time for the task. Since the MTM time values do not include any allowances, the analyst must add the proper allowance according to company policy for the type of work being measured. Having stated this, we are obligated to mention that MTM-1 should not be used in industrial work measurement applications until the analyst has received proper training in the technique.

### 2.2 Other MTM Systems

Additional members of the MTM family have been introduced to satisfy various user needs. Several functional systems have been developed for specific work situations such as clerical activity, machine shop work, and electronic testing. Also, second- and third-level

**TABLE 4 (a)** Normal Time Values for MTM-1 Motion Element: **Reach** (R)

| Distance | | Time in TMU | | | | Hand in Motion | | Case and Description |
|---|---|---|---|---|---|---|---|---|
| cm | inches | A | B | C or D | E | A | B | |
| < 2.0 | < 0.75 | 2.0 | 2.0 | 2.0 | 2.0 | 1.6 | 1.6 | **A** Reach to object in fixed location, or to object in other hand or on which other hand rests. |
| 2.5 | 1 | 2.5 | 2.5 | 3.6 | 2.4 | 2.3 | 2.3 | |
| 5.1 | 2 | 4.0 | 4.0 | 5.9 | 3.8 | 3.5 | 2.7 | |
| 7.6 | 3 | 5.3 | 5.3 | 7.3 | 5.3 | 4.5 | 3.6 | **B** Reach to single object in location that may vary slightly from cycle to cycle. |
| 10.1 | 4 | 6.1 | 6.4 | 8.4 | 6.8 | 4.9 | 4.3 | |
| 12.5 | 5 | 6.5 | 7.8 | 9.4 | 7.4 | 5.3 | 5.0 | |
| 15.2 | 6 | 7.0 | 8.6 | 10.1 | 8.0 | 5.7 | 5.7 | |
| 17.8 | 7 | 7.4 | 9.3 | 10.8 | 8.7 | 6.1 | 6.5 | **C** Reach to object jumbled with other objects in a group so that search and select occur. |
| 20.3 | 8 | 7.9 | 10.1 | 11.5 | 9.3 | 6.5 | 7.2 | |
| 22.9 | 9 | 8.3 | 10.8 | 12.2 | 9.9 | 6.9 | 7.9 | |
| 25.4 | 10 | 8.7 | 11.5 | 12.9 | 10.5 | 7.3 | 8.6 | |
| 30.5 | 12 | 9.6 | 12.9 | 14.2 | 11.8 | 8.1 | 10.1 | **D** Reach to a very small object or where accurate grasp is required. |
| 35.6 | 14 | 10.5 | 14.4 | 15.6 | 13.0 | 8.9 | 11.5 | |
| 40.6 | 16 | 11.4 | 15.8 | 17.0 | 14.2 | 9.7 | 12.9 | |
| 45.7 | 18 | 12.3 | 17.2 | 18.4 | 15.5 | 10.5 | 14.4 | |
| 50.8 | 20 | 13.1 | 18.6 | 19.8 | 16.7 | 11.3 | 15.8 | **E** Reach to indefinite location to get hand in position for body balance or next motion or out the way. |
| 55.9 | 22 | 14.0 | 20.1 | 21.2 | 18.0 | 12.1 | 17.3 | |
| 61.0 | 24 | 14.9 | 21.5 | 22.5 | 19.2 | 12.9 | 18.8 | |
| 66.0 | 26 | 15.8 | 22.9 | 23.9 | 20.4 | 13.7 | 20.2 | |
| 71.1 | 28 | 16.7 | 24.4 | 25.3 | 21.7 | 14.5 | 21.7 | |
| 76.2 | 30 | 17.5 | 25.8 | 26.7 | 22.9 | 15.3 | 23.2 | |
| Additional | | 0.4 | 0.7 | 0.7 | 0.6 | TMU per 2.54 cm > 76 cm (per 1.0 in > 30 in.) | | |

**TABLE 4 (b)** Normal Time Values for MTM-1 Motion Element: **Grasp** (G)

| Type of Grasp | Case | Time, TMU | Description and Object Dimensions | |
|---|---|---|---|---|
| Pickup | 1A | 2.0 | Any size object, by itself | |
| | 1B | 3.5 | Object very small or lying close against a flat surface | |
| | 1C1 | 7.3 | Interference with grasp on bottom and one side of cylindrical object | Diameter > 1.3 cm (0.5 in.) |
| | 1C2 | 8.7 | | Diameter 0.6 to 1.3 cm (0.25 to 0.5 in.) |
| | 1C3 | 10.8 | | Diameter < 0.6 cm (0.25 in.) |
| Regrasp | 2 | 5.6 | Change grasp without relinquishing control | |
| Transfer | 3 | 5.6 | Control transferred from one hand to other | |
| Select | 4A | 7.3 | Object jumbled with other objects so that search and select occur | Size larger than 2.5 × 2.5 × 2.5 cm (1 × 1 × 1 in.) |
| | 4B | 9.1 | | 0.6 × .6 × .3 cm (.25 × .25 × .12 in) to 2.5 × 2.5 × 2.5 cm (1 × 1 × 1 in.) |
| | 4C | 12.9 | | Size smaller than .6 × .6 × .3 cm (.25 × .25 × .12 in.) |
| Contact | 5 | 0 | Contact, sliding, or hook grasp | |

**TABLE 4 (c)** Normal Time Values for MTM-1 Motion Element: **Move** (M)

| Distance | | Time in TMU | | | Hand in motion | Weight up to | Formula Parameters | | Case and Description |
|---|---|---|---|---|---|---|---|---|---|
| cm | inches | A | B | C | B | kg (lb) | Constant | Factor | |
| < 2.0 | < 0.75 | 2.0 | 2.0 | 2.0 | 1.7 | | | | **A** Move object to other hand or against stop. |
| 2.5 | 1 | 2.5 | 2.9 | 3.4 | 2.3 | 1.1 (2.5) | 0 | 1.00 | |
| 5.1 | 2 | 3.6 | 4.6 | 5.2 | 2.9 | | | | |
| 7.6 | 3 | 4.9 | 5.7 | 6.7 | 3.6 | 3.4 (7.5) | 2.2 | 1.06 | |
| 10.1 | 4 | 6.1 | 6.9 | 8.0 | 4.3 | | | | **B** Move object to approximate or indefinite location. |
| 12.5 | 5 | 7.3 | 8.0 | 9.2 | 5.0 | 5.7 (12.5) | 3.9 | 1.11 | |
| 15.2 | 6 | 8.1 | 8.9 | 10.3 | 5.7 | | | | |
| 17.8 | 7 | 8.9 | 9.7 | 11.1 | 6.5 | 7.9 (17.5) | 5.6 | 1.17 | |
| 20.3 | 8 | 9.7 | 10.6 | 11.8 | 7.2 | | | | |
| 22.9 | 9 | 10.5 | 11.5 | 12.7 | 7.9 | 10.2 (22.5) | 7.4 | 1.22 | **C** Move object to exact location. |
| 25.4 | 10 | 11.3 | 12.2 | 13.5 | 8.6 | | | | |
| 30.5 | 12 | 12.9 | 13.4 | 15.2 | 10.0 | 12.5 (27.5) | 9.1 | 1.28 | |
| 35.6 | 14 | 14.4 | 14.6 | 16.9 | 11.4 | | | | |
| 40.6 | 16 | 16.0 | 15.8 | 18.7 | 12.8 | 14.7 (32.5) | 10.8 | 1.33 | |
| 45.7 | 18 | 17.6 | 17.0 | 20.4 | 14.2 | | | | |
| 50.8 | 20 | 19.2 | 18.2 | 22.1 | 15.6 | 17.0 (37.5) | 12.5 | 1.39 | |
| 55.9 | 22 | 20.8 | 19.4 | 23.8 | 17.0 | | | | |
| 61.0 | 24 | 22.4 | 20.6 | 25.5 | 18.4 | 19.3 (42.5) | 14.3 | 1.44 | |
| 66.0 | 26 | 24.0 | 21.8 | 27.3 | 19.8 | | | | |
| 71.1 | 28 | 25.5 | 23.1 | 29.0 | 21.2 | 21.5 (47.5) | 16.0 | 1.50 | |
| 76.2 | 30 | 27.1 | 24.3 | 30.7 | 22.7 | | | | |
| Additional | | 0.8 | 0.6 | 0.85 | TMU per 2.54 cm > 76 cm (per 1.0 in. > 30 in.) | | | | |

**TABLE 4 (d)** Normal Time Values for MTM-1 Motion Element: **Position** (P)

| | | | Time in TMU | |
|---|---|---|---|---|
| Class | Description of Fit | Symmetry | Easy to Handle | Difficult to Handle |
| 1 | Loose (no pressure required) | S | 5.6 | 11.2 |
| | | SS | 9.1 | 14.7 |
| | | NS | 10.4 | 16.0 |
| 2 | Close (light pressure required) | S | 16.2 | 21.8 |
| | | SS | 19.7 | 25.3 |
| | | NS | 21.0 | 26.6 |
| 3 | Exact (heavy pressure required) | S | 43.0 | 48.6 |
| | | SS | 46.5 | 52.1 |
| | | NS | 47.8 | 53.4 |
| *Key*: S = symmetrical, SS = semi-symmetrical, NS = nonsymmetrical. | | | | |

**TABLE 4 (e)** Normal Time Values for MTM-1 Motion Element: **Release** (RL)

| Case | Time in TMU | Description |
|---|---|---|
| 1 | 2.0 | Normal release performed by opening fingers as an independent motion |
| 2 | 0 | Contact release with no finger motion |

**TABLE 4 (f)** Normal Time Values for MTM-1 Motion Element: **Disengage** (D)

| | | | Time in TMU | |
|---|---|---|---|---|
| Class | Description of Fit | Height of Recoil | Easy to Handle | Difficult to Handle |
| 1 | Loose (very slight effort, blends with subsequent move) | Up to 2.5 cm (1 in) | 4.0 | 5.7 |
| 2 | Close (normal effort, slight recoil) (1 to 5 in) | 2.5 to 12.7 cm | 7.5 | 11.8 |
| 3 | Tight (considerable effort, hand recoils markedly) | 12.7 to 30 cm (5 to 12 in) | 22.9 | 34.7 |

**TABLE 4 (g)** Normal Time Values for MTM-1 Motion Element: **Turn** (T)

| | Time in TMU for Degrees Turned | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Weight, kg (lb) | 30° | 45° | 60° | 75° | 90° | 105° | 120° | 135° | 150° | 165° | 180° |
| Small, up to 0.9 (2) | 2.8 | 3.5 | 4.1 | 4.8 | 5.4 | 6.1 | 6.8 | 7.4 | 8.1 | 8.7 | 9.4 |
| Medium, 1 to 4.5 (2 to 10) | 4.4 | 5.5 | 6.5 | 7.5 | 8.5 | 9.6 | 10.6 | 11.6 | 12.7 | 13.7 | 14.8 |
| Large, 4.5 to 16 (10 to 35) | 8.4 | 10.5 | 12.3 | 14.4 | 16.2 | 18.3 | 20.4 | 22.2 | 24.3 | 26.1 | 28.2 |

**TABLE 4 (h)** Normal Time Values for MTM-1 Motion Element: **Apply Pressure** (AP)

| Symbol | Time in TMU | Description |
|---|---|---|
| APA | 10.6 | Apply pressure alone |
| APB | 16.2 | Apply pressure preceded by regrasp |

**TABLE 4 (i)** Normal Time Values for MTM-1 Motion Element: **Eye Travel** (ET) and **Eye Focus** (EF)

| Eye motion | Symbol | Time in TMU | Key to Symbols |
|---|---|---|---|
| Eye travel | ET | $\frac{15.2L}{D}$ | $L$ = distance between points from and to which eye travels, $D$ = perpendicular distance from the eye to the line of travel. Maximum time allowed = 20 TMU |
| Eye focus | EF | 7.3 | |
| Reading | (none) | $5.05N$ | $N$ = number of words read (330 words/min) |

**TABLE 4 (j)** Normal Time Values for MTM-1 Motion Element: **Body, leg**, and **foot motions** (various symbols given in table)

| Motion | Symbol | Time in TMU | Description and Conditions |
|---|---|---|---|
| Sit | SIT | 34.7 | From standing position |
| Stand | STD | 43.4 | From seated position |
| Turn body | TBC1 | 18.6 | Turn body 45° to 90°, Case 1 – Lagging foot not aligned with leading foot |
| Turn body | TBC2 | 37.2 | Turn body 45° to 90°, Case 2 – Lagging foot aligned with leading foot |
| Bend | B | 29.0 | Bend body forward so hands can reach knees |
| Stoop | S | 29.0 | Stoop body forward so hands can reach floor |
| Arise | AB | 31.9 | Arise from bent position |
| Arise | AS | 31.9 | Arise from stooped position |
| Kneel | KOK | 29.0 | Kneel on one knee |
| Kneel | KBK | 69.4 | Kneel on both knees |
| Arise | AKOK | 31.9 | Arise from kneeling position on one knee |
| Arise | AKBK | 76.7 | Arise from kneeling position on both knees |
| Walk | W*X*FT | 5.3 per ft | Walking in ft of distance, $X$ = distance in ft |
| Walk | W*N*P | 15.0/pace | Walking in number of paces, $N$ = number of paces |
| Walk | W*N*PO | 17.0/pace | Walking in number of paces with weight or obstruction, $N$ = number of paces |
| Leg motion | LM6 | 7.1 | Move leg up to 6 in. any direction |
| Leg motion | LM*X* | 7.1 + 1.2($X$-6) | Move leg more than 6 in. any direction, where $X$ = distance of movement |
| Foot motion | FM | 8.5 | Foot moves up to 4 in. hinged at ankle |
| Foot motion | FMP | 19.1 | Foot moves up to 4 in. hinged at ankle, apply heavy pressure with leg muscles |

**TABLE 4 (k)** MTM-1 Simultaneous Hand and Arm Motion Elements

| Motion of One Hand | | Reach | | | Grasp | | | Move | | | Position | | | Disengage | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Motion of other hand | Case or Class | A E | B | C D | 1A 2 5 | 1B 1C | 4 | A Bm[a] | B | C | 1S | 1SS 2S | 1NS 2SS 2NS | 1E 1D | 2 |
| Reach | A, E | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 2 | 1 | 1 | 2 | 1 | 1 |
| | B | 1 | 1 | 1 | 1 | 1 | 2 | 1 | 1 | 2 | 2 | 2 | 3 | 1 | 1 |
| | C, D | 1 | 1 | 1 | 1 | 2 | 3 | 2 | 2 | 3 | 3 | 3 | 3 | 2 | 3 |
| Grasp | 1A, 2, 5 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 3 | 1 | 3 |
| | 1B, 1C | 1 | 1 | 2 | 1 | 3 | 2 | 1 | 1 | 2 | 3 | 3 | 3 | 3 | 3 |
| | 4 | 1 | 2 | 3 | 1 | 2 | 3 | 1 | 2 | 3 | 3 | 3 | 3 | 3 | 3 |
| Move | A | 1 | 1 | 2 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 2 | 1 | 1 |
| | B | 1 | 1 | 2 | 1 | 1 | 2 | 1 | 1 | 1 | 2 | 2 | 3 | 1 | 1 |
| | C | 2 | 2 | 3 | 1 | 2 | 3 | 1 | 1 | 2 | 3 | 3 | 3 | 2 | 3 |
| Position | 1S | 1 | 2 | 3 | 1 | 3 | 3 | 1 | 2 | 3 | 2 | 3 | 3 | 3 | 3 |
| | 1SS,2S | 1 | 2 | 3 | 1 | 3 | 3 | 1 | 2 | 3 | 3 | 3 | 3 | 3 | 3 |
| 1NS, 2SS,2NS | | 2 | 3 | 3 | 3 | 3 | 3 | 2 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| Disengage | 1E,1D | 1 | 1 | 2 | 1 | 3 | 3 | 1 | 1 | 2 | 3 | 3 | 3 | 1 | 1 |
| | 2 | 1 | 1 | 3 | 3 | 3 | 3 | 1 | 1 | 3 | 3 | 3 | 3 | 1 | 1 |

[a]Bm is Case B with hand in motion.

(*continued*)

**TABLE 4 (k)** (*continued*)

*Key:* The cell numbers indicate the degree of difficulty when motions are performed simultaneously.
1 = Easy to perform simultaneously. Use the longest motion element time.
2 = Can be performed simultaneously with practice. Use the longest motion element time.
3 = Difficult to perform simultaneously. Add the times of the two simultaneous motion elements.

*Assumptions:* All Reach, Grasp, and Move motions are performed within the area of normal vision. In the Position and Disengage motion elements, objects are assumed easy to handle. In general, the degree of difficulty increases if these assumptions are violated.

*Motions not included in the table:*
Turn: normally degree of difficulty = 1 except when Turn is controlled or with Disengage.
Position Class 3: degree of difficulty = 3.
Disengage Class 3: normally degree of difficulty = 3.
Release: degree of difficulty = 1.

**TABLE 5** Members of the MTM Family of Predetermined Motion Time Systems

**MTM-1**: The first level MTM, in which basic motion elements are used to describe, analyze, and determine the normal time for a manual task. The MTM-1 motion elements and associated times are listed in Table 3 and 4. MTM-1 is best suited to high production operations with relatively short cycle times. The analyst time required to apply MTM-1 is about 250 times the task cycle time.

**MTM-2**: A second-level MTM system, in which the basic motion elements are combined into motion aggregates in order to reduce the analyst's time to apply the technique. MTM-2 consists of 11 motions and motion aggregates, called *motion categories*. The two most important MTM-2 categories are GET, which combines Reach, Grasp, and Release; and PUT, which combines Move and Position. MTM-2 is suited to operations that are not highly repetitive and the cycle times are greater than 1 min. The analyst time required to apply MTM-2 is about 100 times the task cycle time.

**MTM-3**: A third-level MTM system designed to further reduce the analyst's time to set a time standard but at some sacrifice in accuracy. MTM-3 has only four motion categories: (1) Handle, (2) Transport, (3) Step and foot motions, and (4) Bend and arise. Tasks that include additional elements, such as eye movements, should not be analyzed using MTM-3. The analyst time required to apply MTM-3 is about 35 times the task cycle time.

**MTM-UAS**: A third-level MTM system that is suited to applications in batch production. It includes seven basic motion categories: (1) Get and place, (2) Place, (3) Handle tool, (4) Operate, (5) Motion cycles, (6) Body motions, and (7) Visual control. Additional second-level standard data in MTM-UAS cover activities such as fastening, marking, packing, and assembling. The analyst time required to apply MTM-UAS is about 30 times the task cycle time.

**MTM-MEK**:A third-level system intended for work measurement applications in small lot production with long cycle times and other tasks performed infrequently. Time standards are usually not established in these work situations by conventional time study because the cost of setting the standard is too high. MTM-MEK can be used for these cases and the analyst time is about 5 to 15 times the cycle time.

**MTM-HC**: A functional PMTS designed for work activities found in the health-care industry. It is described as a standard database by the MTM Association.

**MTM-C**: A functional MTM system used for work measurement applications of clerical work activity, such as typing or keypunching, filing, reading, and writing. It was developed by an association of banking and other service industries. There are two levels of MTM-C: MTM-C1 and MTM-C2. The difference is that MTM-C1 emphasizes precision and motion detail in its applications, while MTM-C2 emphasizes speed of application. The analyst time to apply MTM-C1 is about 125 times the task cycle time, and about 75 times for MTM-C2.

**MTM-V**: A functional standard data system developed using MTM-1 for work measurement in machine tool operation. Its work elements include handling of work parts, operating a machine tool and other equipment (e.g., cranes,

(*continued*)

**TABLE 5** (*continued*)

fixtures, chucks), and setting up a job for production. The analyst time required to apply MTM-V is about 10 times the task cycle time.

**MTM-TE**: A functional standard data system designed for work measurement in electronic testing applications. Its work elements cover both manual and mental activities.

**MTM-M**: A functional MTM system for measuring assembly work that is performed under a stereoscopic microscope.

*Source:* Estimates of analyst time to apply the various MTM systems are based on [7].

systems are available to reduce the time required to develop time standards. Finally, several computerized systems have been developed based on MTM. Table 5 lists many of these MTM systems with a brief description of each. Readers interested in learning more about the other MTM systems can consult the coverage provided in [6] or the MTM Web site at www.mtm.org.

## 3 MAYNARD OPERATION SEQUENCE TECHNIQUE

The Maynard Operation Sequence Technique (MOST) is a high-level predetermined motion time system that is based on MTM. It uses the same time units as in MTM: the TMU (time measurement unit), which is 0.00001 hr. MOST is a product of H. B. Maynard and Company, an educational and consulting firm located in Pittsburgh, Pennsylvania.[4] The basic version of MOST, which is now referred to as Basic MOST, was developed at Maynard's Swedish Division around 1967 under the direction of Kjell Zandin (Historical Note 1). Section 3.1, describes the Basic MOST package. Several additional versions of MOST have been developed since Basic MOST was introduced, and these are summarized in Section 3.2. Section 3.3 describes a computerized version of MOST.

### 3.1 Basic MOST

The focus of Basic MOST is on work activity involving the movement of objects. In fact, the majority of industrial manual work does involve moving objects (e.g., parts, tools) from one location to another in the workplace. The underlying premise of Basic MOST is that the movement of objects consists of a pattern of body motions and actions that is nearly the same for all moves. Only the details of the pattern differ, depending on the object being moved and the circumstances of the work activity. Consistent with this premise, Basic MOST uses motion aggregates (collections of basic motion elements) that are concerned with moving things. The motion aggregates are called ***activity sequence models*** in Basic MOST. In our hierarchy of work activity, an activity sequence model is either at the same level as a work element, or slightly below. Depending on how a work element is defined for a given task, it may include one or more activity sequence models. Rarely would a work element be defined as less than a complete activity sequence model. The relationships are indicated in Figure 2.

[4]MOST is a registered trademark of H. B. Maynard and Company, Inc.

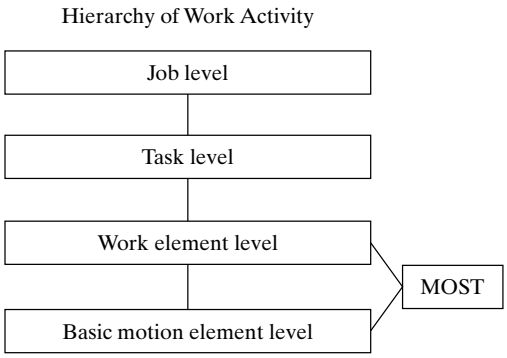



**Figure 2** The position of the Basic MOST activity sequence model in our work hierarchy.

There are three activity sequence models in Basic MOST, each of which consists of a standard sequence of actions:

- *General move.* This sequence model is used when an object is moved freely through space from one location to the next (e.g., picking something up from the floor and placing it on a table).
- *Controlled move.* This sequence model is used when an object is moved while it remains in contact with a surface (e.g., sliding the object along the surface) or the object is attached to some other object during its movement (e.g., moving a lever on a machine).
- *Tool use.* This sequence model applies to the use of a hand tool (e.g., a hammer or screwdriver).

The actions in an activity sequence model, called ***sequence model parameters*** in Basic MOST, are similar to basic motion elements in MTM, but there is not a one-to-one correspondence. Some parameters map directly into MTM basic motions, while others are unique to Basic MOST. Let us examine the three sequence models and indicate the standard sequence of model parameters for each.[5]

**General Move.** The General Move sequence is applicable when an object is moved through the air from one location to another. There are four parameters (actions) in the General Move, symbolized by letters of the alphabet:

A—Action distance, usually horizontal. This parameter is used to describe movements of the fingers, hands, or feet (e.g., walking). The movement can be performed either loaded or unloaded.

B—Body motion, usually vertical. This parameter defines vertical body motions and actions (e.g., sitting, standing up).

[5]Once again, we must caution against the application of MOST without proper training. Guidelines can be obtained from H. B. Maynard & Company in Pittsburgh, Pennsylvania; their Web site can be accessed at www.hbmaynard.com.

G—Gain control. This parameter is used for any manual actions involving the fingers, hands, or feet to gain physical control of one or more objects. It is closely related to the grasp motion element in MTM (e.g., grasp the object).

P—Placement. The placement parameter is used to describe the action involved to lay aside, position, orient, or align an object after it has been moved to the new location (e.g., position the object).

These parameters occur in the following standard sequence in the General Move:

| A | B | G | A | B | P | A |
|---|---|---|---|---|---|---|
| Action distance | Body motion | Grasp | Action distance | Body motion | Placement | Action distance |

where the first three parameters (A B G) represent basic motions to ***get*** an object, the next three parameters (A B P) represent motions to ***put*** or move the object to a new location, and the final parameter (A) applies to any motions at the end of the sequence, such as ***return*** to original position.

To complete the activity sequence model, each parameter is assigned a numerical value in the form of a subscript or index number that represents the time to accomplish that action. The value of the index number depends on the type of action, its motion content, and the conditions under which it is performed. Table 6 lists the parameters and possible circumstances for the action, together with the corresponding values of the index numbers. When the index values have been entered for all parameters, the time for the sequence model is determined by summing the index values and multiplying by 10 to obtain the total TMUs. The procedure is illustrated in the following example.

**Example 1 General Move**

Develop the activity sequence model and determine the normal time for the following work activity: A worker walks 5 steps, picks up a small part from the floor, returns to his original position, and places the part on his worktable.

**Solution:** Referring to Table 6, the indexed activity sequence model for this work activity would be the following:

$$A_{10}B_6G_1A_{10}B_0P_1A_0$$

where $A_{10}$ = walk 5 steps, $B_6$ = bend and arise, $G_1$=gain control of small part, $A_{10}$ = walk back to original position, $B_0$ = no body motion, $P_1$ = lay aside part on table, and $A_0$ = no motion. The sum of the index values is 28. Multiplying by 10, we have 280 TMUs (about 10 sec). ■

The General Move is the most common way in which materials are moved and is therefore the most frequently used activity sequence model in Basic MOST. Approximately half of all manual work occurs as General Moves. For assembly work and material handling, the proportion is higher. For machine shop work, the proportion is lower.

**TABLE 6** MOST Parameters and Index Values for the General Move Activity Sequence Model

| General Move activity sequence model = **A B G A B P A** | | | | |
|---|---|---|---|---|
| Index | **A** = Action distance | **B** = Body motion | **G** = Gain control | **P** = Placement |
| **0** | Close ≤ 5 cm (2 in.) | | | Hold, Toss |
| **1** | Within reach (but > 2 in.) | | Grasp light object using one or two hands | Lay aside Loose fit |
| **3** | 1 or 2 steps | Bend and arise with 50% occurrence | Grasp object that is heavy, or obstructed, or hidden, or interlocked | Adjustments, light pressure, double placement |
| **6** | 3 or 4 steps | Bend and arise with 100% occurrence | | Position with care, or precision, of blind, or obstructed, or heavy pressure |
| **10** | 5, 6, or 7 steps | Sit or stand | | |
| **16** | 8, 9, or 10 steps | Through door, or Climb on or off, or Stand and bend, or Bend and sit | | |

**Controlled Move.** The Controlled Move sequence model is used when an object is moved through a path that is somehow constrained. The object cannot be freely moved through the air. This situation arises when an object is slid across a surface or when the object is attached to something else and it can be moved only through a controlled path. The second case is commonly encountered in the operation of machinery, and the move involves a lever, crank, switch button, or similar entity that is attached to the machinery. The parameters in the Controlled Move include the A, B, and G parameters from the General Move. In addition, three new parameters are introduced:

M—Move, controlled. This parameter is used to describe any manual body motions required to move an object over a controlled path.

X—Process time. Since the Controlled Move can include the operation of machinery, there is often a process time associated with the machinery (e.g., time to turn the workpiece in a lathe). No manual motions are included in this parameter.

I—Align. This parameter is used when manual motions are performed at the end of the controlled move to align objects.

These parameters occur in the following standard sequence in the Controlled Move:

| A | B | G | M | X | I | A |
|---|---|---|---|---|---|---|
| Action distance | Body motion | Grasp | Move, controlled | Process time | Align | Action distance |

where the first three parameters (A B G) are the basic motions to ***get*** an object, the next three parameters (M X I) represent moving or actuating the object followed by a process time and alignment, and the final parameter (A) provides for a possible ***return*** at the end of the sequence.

**TABLE 7** MOST Parameters and Index Values for the Controlled Move Activity Sequence Model

| Controlled Move activity sequence model = **A B G M X I A** | | | | |
|---|---|---|---|---|
| Index | **M** = Move, controlled | **X** = Process time[a] | | **I** = Alignment |
| | | Seconds | Minutes | |
| **1** | Push, pull, pivot: button, switch, knob (≤ 12 in.) | 0.5 | 0.01 | Align to one point |
| **3** | Push and pull, turn, open, seat, shift, press: resistance encountered, or high control required, or 2 stages of control (≤ 12 in.); 1 crank of lever. | 1.5 | 0.02 | Align to 2 points, Close align (≤ 4 in.) |
| **6** | Open and shut, operate, push or pull: with 1 or 2 steps (> 12 in.); 3 cranks of lever. | 2.5 | 0.04 | Align to 2 points, Close align (> 4 in.) |
| **10** | Manipulate, maneuver, push, or pull with 3, 4, or 5 steps; 6 cranks of lever. | 4.5 | 0.07 | Precision align |
| **16** | Push or pull with 6, 7, 8, or 9 steps included; 11 cranks of lever. | 7.0 | 0.11 | High precision align |

[a]For process times longer than those listed in the table, the actual process time in seconds can be multiplied by 2.78 and rounded to the next higher value to obtain the index for the X parameter.

Index values are applied to each parameter as in the General Move. The same A, B, and G index values from Table 6 apply to the Controlled Move. Table 7 lists the index values for the M, X, and I parameters used exclusively for the Controlled Move.

**Example 2 Controlled Move**

Develop the activity sequence model and determine the normal time for the following work activity: A worker takes 2 steps, grasps the waist-level feed lever on the lathe, pulls up the lever approximately 15 cm to engage the feed. The process time to turn the part is 25 sec. There is no alignment and no action by the worker at the end of the process time.

**Solution:** Referring to Tables 6 and 7, and multiplying the 25 sec process time by 2.78 to obtain a value of 69.5 for the X parameter (rounded up to 70), we obtain the following indexed activity sequence model for this work activity:

$$A_3\ B_0\ G_1\ M_1\ X_{70}\ I_0\ A_0$$

where $A_3$ = take 2 steps, $B_0$ = no body motion, $G_1$ = gains control of lever, $M_1$ = pull the lever up 15 cm, $X_{70}$ = process time of 25 sec, $I_0$ = no alignment, and $A_0$ = no motion. The sum of the index values is 75. Multiplying by 10, we have 750 TMUs (about 27 sec). The lathe process time accounts for all but 2 sec of the activity sequence. ■

**Tool Use.** The Tool Use sequence model applies to the variety of work situations that involve the use of a hand tool. Hand tools covered by this model include screwdrivers, ratchets, hammers, scissors, knives, various kinds of wrenches, and common measuring instruments. When a worker's hand is used to perform a motion such as turning a screw, Tool Use is employed. The sequence model also covers writing, marking, reading, thinking, and inspecting. The parameters in the Tool Use sequence include A, B, G, and

P that were previously used in the General Move and Controlled Move. Only one new parameter is included in Tool Use, and that refers to the specific action that applies to the tool being used in the activity sequence. The possible parameters include the following:

F—Fasten. This parameter is used to describe any fastening motion that is performed with the fingers, hand, or a tool such as a screwdriver or wrench.

L—Loosen. This parameter is similar to Fasten, except the objective is to unfasten or disassemble an object.

C—Cut. The Cut parameter describes the manual actions required to slice, slit, or otherwise separate an object. Common tools for cutting include knives, scissors, and pliers.

S—Surface treat. This parameter is used for applying a coating to the surface of an object (e.g., using a brush) or to remove a material from the surface (e.g., wiping the surface).

M—Measure. The use of measuring instruments (e.g., linear scales, calipers, micrometers) or gauges is modeled by this parameter.

R—Record. The manual actions involving the use of a pen, pencil, or other marking tool to record data are covered by this parameter.

T—Think. This parameter models the visual and mental activities required to read or inspect an object.

Only one of these parameters is included in the standard sequence for the Tool Use activity sequence model. Its placement in the sequence model is indicated by the asterisk (*) in the following:

A B G A B P * A B P A

where the first three parameters (A B G) are the motions to ***get*** the tool, the next three parameters (A B P) ***put*** or place the tool into position, * is the tool use code (F, L, C, S, M, R, or T), the next three parameters (A B P) ***put*** the tool aside, and the final parameter (A) is to provide for the worker to ***return*** to some previous position. Table 8 (a) and (b) list the index values for the Tool Use parameters.

**Example 3 Tool Use**

Develop the activity sequence model and determine the normal time for the following work activity: A worker picks up a screw from his worktable, positions it into a threaded hole, and turns it three spins with his fingers. He then picks up a screwdriver from the worktable, positions it on the head of the screw, fastens the screw with six turns, and lays the screwdriver aside.

**Solution:** This requires two activity sequence models, corresponding to the two sentences of the work description. The first model is developed as follows:

$$A_1\ B_0\ G_1\ A_1\ B_0\ P_3\ F_6\ A_0\ B_0\ P_0\ A_0$$

**TABLE 8 (a)** MOST Parameters and Index Values for Tool Use—Fasten and Loosen

| Fasten or Loosen = **F** or **L** | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Finger Action | Wrist Action | | | | Arm Action | | | | | Tool Action | |
| Index | Fingers, Screw-driver | Hand, Screw-driver, Ratchet, T-wrench | Wrench, Allen key | Wrench, Allen key, Ratchet | Hand, Hammer | Ratchet | T-wrench, 2 hands | Wrench, Allen key | Wrench, Allen key, Ratchet | Hand, Hammer | Power wrench | Index |
| | Spins | Turns | Strokes | Cranks | Taps | Turns | Turns | Strokes | Cranks | Strikes | Diameter | |
| **1** | 1 | — | — | — | 1 | — | — | — | — | — | — | **1** |
| **3** | 2 | 1 | 1 | 1 | 3 | 1 | — | 1 | — | 1 | 6 mm (1/4 in.) | **3** |
| **6** | 3 | 3 | 2 | 3 | 6 | 2 | 1 | — | 1 | 3 | 25 mm (1 in.) | **6** |
| **10** | 8 | 5 | 3 | 5 | 10 | 4 | — | 2 | 2 | 5 | | **10** |
| **16** | 16 | 9 | 5 | 8 | 16 | 6 | 3 | 3 | 3 | 8 | | **16** |
| **24** | 25 | 13 | 8 | 11 | 23 | 9 | 6 | 4 | 5 | 12 | | **24** |
| **32** | 35 | 17 | 10 | 15 | 30 | 12 | 8 | 6 | 6 | 16 | | **32** |
| **42** | 47 | 23 | 13 | 20 | 39 | 15 | 11 | 8 | 8 | 21 | | **42** |
| **54** | 61 | 29 | 17 | 25 | 50 | 20 | 15 | 10 | 11 | 27 | | **54** |

**TABLE 8 (b)** Parameters and Index Values for Tool Use—Cut (C), Surface treat (S), Measure (M), Record (R), and Think (T)

| | | Cut = **C** | | | Surface Treat = **S** | | | Measure = **M** | Record = **R** | | Think = **T** | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Cutoff | Secure | Cut | Slice | Air clean | Brush clean | Wipe | Measure | Write | | Mark | Inspect | Read | | |
| | Pliers | | Scissors | Knife | Nozzle | Brush | Cloth | Measuring Tool | Pen/Pencil | | Marker | Eyes, fingers | Eyes | | |
| Index | Wire | | Cuts | Slices | .1 $m^2$ (1 $ft^2$) | .1 $m^2$ (1 $ft^2$) | .1 $m^2$ (1 $ft^2$) | | Digits | Words | Digits | Points | Digits, single words | Text | Index |
| **1** | | Grip | 1 | | | | | | 1 | | Check | 1 | 1 | 3 | **1** |
| **3** | Soft | | 2 | 1 | | | 1/2 | | 2 | | 1 Scribe line | 3 | 3 | 8 | **3** |
| **6** | Medium | Twist Form loop | 4 | | 1 Spot cavity | 1 | | | 4 | 1 | 2 | 5 Feel for heat | 6 | 15 | **6** |
| **10** | Hard | | 7 | 3 | | | 1 | Profile gauge | 6 | | 3 | 9 Feel defect | 12 | 24 | **10** |
| **16** | | Secure cotter pin | 11 | 4 | 3 | 2 | 2 | Fixed scale, Caliper | 9 | 2 | 5 | 14 | | 38 | **16** |
| **24** | | | 15 | 6 | 4 | 3 | | Feeler gauge | 13 | 3 | 7 | 19 | | 54 | **24** |
| **32** | | | 20 | 9 | 7 | 5 | 5 | Steel tape, Depth micrometer | 18 | 4 | 10 | 26 | | 72 | **32** |
| **42** | | | 27 | 11 | 10 | 7 | 7 | OD micrometer | 23 | 5 | 13 | 34 | | 94 | **42** |
| **54** | | | 33 | | | | | ID micrometer | 29 | 7 | 16 | 42 | | 119 | **54** |

where $A_1\ B_0\ G_1$ = pick up screw from worktable, $A_1\ B_0\ P_3$ = position screw into threaded hole, $F_6$ = turn screw three spins with fingers, and $A_0\ B_0\ P_0\ A_0$ = no motions. The second activity sequence model is the following:

$$A_1\ B_0\ G_1\ A_1\ B_0\ P_3\ F_{10}\ A_1\ B_0\ P_1\ A_0$$

where $A_1\ B_0\ G_1$ = pick up screwdriver from worktable, $A_1\ B_0\ P_3$ = position screwdriver on screw head, $F_{10}$ = turn screwdriver six turns, $A_1\ B_0\ P_1$ = lay screwdriver aside, and $A_0$ = no motion.

The sum of the index values is 12 for the first model and 18 for the second, a total of 30. Multiplying by 10, we have 300 TMUs (about 10.8 sec). ■

### 3.2 Additional Versions of MOST

In addition to Basic MOST, several additional versions of MOST are available for other application characteristics. These are described in Table 9. A description of Basic MOST is included for comparison purposes.

### 3.3 MOST for Windows

Keeping up with trends in work measurement, H. B. Maynard and Company has developed a computerized version of MOST, called MOST for Windows. Depending on the application, MOST for Windows allows the user to analyze a given work situation by any of the three MOST analysis techniques: (1) Basic MOST, (2) Maxi MOST, or (3) Mini

**TABLE 9** Other Versions of MOST Work Measurement Packages

**Basic MOST**: A system designed for manual work that involves the movement of objects. It is intended for applications in which the work cycle is performed more than 150 times per week but less than 1500 times per week. A typical cycle time for Basic MOST is 30 sec to 3 min. As described in Section 3.1, Basic MOST uses three activity sequence models: (1) General Move, (2) Controlled Move, and (3) Tool Use. The analyst time required to apply Basic MOST is about 10 times the task cycle time.

**Maxi MOST**: A system developed for applications in which the work cycle is performed fewer than 150 times per week and there are variations in the cycle. Task times of several hours can be analyzed using this version. Examples of work activity that can be analyzed by Maxi MOST include setup tasks and heavy assembly. Maxi MOST consists of five activity sequence models: (1) part handling, (2) tool/equipment use, (3) machine handling, (4) transport with powered crane, and (5) transport with wheeled truck. The analyst time required to apply Maxi MOST is 3 to 5 times the task cycle time.

**Mini MOST**: A system developed for operations in which the work cycle is highly repetitive and performed more than 1500 times per week. For a 40-hour week, this translates to a cycle time of 1.6 min or less. However, cycle times analyzed by Mini MOST are usually much less than this. Mini MOST consists of two activity sequence models: (1) General Move and (2) Controlled Move. The analyst time required to apply Mini MOST is about 25 times the task cycle time.

**Clerical MOST**: A system similar to Basic MOST but designed for clerical work activity. In addition to the three basic activity sequence models in Basic MOST (General Move, Controlled Move, and Tool Use), there is an additional model for equipment use. The format of the model is the same as Tool Use, except that additional parameters are available for actions involving the operation of office equipment.

*Source:* Estimates of analyst time to apply the MOST versions are based on [9].

MOST (Table 9). Other alternatives are also available. The analyst can use action distances (the A parameter in Basic MOST) and body motions (the B parameter) that are based on data about the layout of the workplace, rather than determining and inputting the index values directly.

Among the other options, MOST for Windows includes two modules that can be called by the analyst:

- *Quick MOST*. This feature permits the user to select among multiple possible work methods that might be applied to the given application. The method description, activity sequence model, and associated normal time are immediately provided by Quick MOST, where editing can be performed by the analyst if needed. The selection list of work methods is based on data of industrial applications compiled by Maynard. The user can add additional work methods representing the company's unique applications if desired.
- *Direct MOST*. This MOST for Windows feature automatically generates a description of the work method based on data provided about the work situation. The module allows the analyst to override or edit the automatically generated work method description when appropriate. Parameter index values are automatically developed by Direct MOST when the analyst selects rules that have been developed for MOST applications. These rules are available to the user through drop-down menus, or by specification of workplace objects and locations.

The following are the benefits and advantages reported by Maynard for users of MOST for Windows: (1) A complete methods description is provided. (2) Nonproductive work elements are identified during the analysis so that they can be eliminated from the standard method. (3) MOST for Windows is a simple and easy-to-use work measurement technique. (4) Developing time standards by MOST for Windows is efficient and accurate. (5) Database search capabilities are provided.

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